CSC4197

RESEARCH METHODOLOGY

THESIS WRITING & RESEARCH DISSEMINATION



2 THESIS VS. PAPER: WHAT CHANGES?

Aspect	Conference paper	Undergraduate thesis
Length	4–14 pages	40–100+ pages
Audience	Expert reviewers	Examiners incl. non-specialists
Detail	Compressed	Everything shown: derivations, full results, appendices
Literature review	0.5–1.5 pages	A full chapter
Voice	"We"	Often "we"/passive per university guide

A thesis must **teach**; a paper must **convince**.

3 STANDARD THESIS STRUCTURE

- **Front matter:** title page → declaration → approval page → abstract → acknowledgments → ToC → list of figures/tables → abbreviations
- **Chapters (typical CS):**
  - Introduction (problem, RQs, contributions, organization)
  - Literature Review / Background
  - Methodology (design, data, tools)
  - Implementation / System Design *(for DSR work)*
  - Results & Analysis
  - Discussion (interpretation, threats to validity)
  - Conclusion & Future Work
- **Back matter:** references → appendices (instruments, extra tables, code listings)
- Always follow **your department's thesis template** — formatting compliance is checked before content.

4 THESIS PROJECT MANAGEMENT

- **Backward-plan from the defense date:** defense ← submission ← supervisor sign-off ← full draft ← chapter drafts ← experiments done
- Write **as you go** — methodology chapter while building, not after
- **Supervisor etiquette:** regular short meetings with agenda; send drafts ≥1 week before you need feedback; maintain a decision log; never disappear for a month
- Version everything (Overleaf/Git); back up data in **3 places**
- The thesis is a marathon: 500 words/day beats 5,000-word panic weekends

5 THEORY: DISSEMINATION CHANNELS

- Where research results travel:
- **Conferences** — fast, deadline-driven, networking; CS prestige venues
- **Journals** — deeper review, no deadline, archival weight
- **Workshops** — early ideas, friendly feedback, community entry point
- **Preprints (arXiv)** — instant visibility, citable, not peer-reviewed
- **Posters & demos** — conversations, recruiting collaborators
- **Theses repositories**, institutional archives
- **Research communication:** blogs, LinkedIn/X threads, talks — growing impact channel
- A typical arc: workshop → conference → journal extension (+~30% new material).

6 CHOOSING A VENUE

- Checklist:
- **Scope fit** — read the CFP topics; check last year's accepted titles
- **Quality tier** — CORE rank / Scimago quartile vs. your work's maturity
- **Timeline** — deadline, notification, camera-ready vs. your graduation date
- **Cost** — registration, APC for open access; fee waivers exist
- ⚠️ **Predatory check** — Think-Check-Submit: Do you/colleagues know the venue? Real editorial board? Indexed where claimed? Suspiciously fast "review"? Flattering spam invitation = red flag
- Realistic undergrad targets: national/regional IEEE conferences, student tracks, workshops.

7 THE PEER-REVIEW PIPELINE

- Submit (EasyChair/CMT/OpenReview) → desk check → 2–4 reviewers
- → scores + comments → decision:
- Accept | Minor revision | Major revision | Reject
- → (journals) revise + response letter → re-review → ...
- → camera-ready → publication
- Review criteria: novelty, soundness, significance, clarity, reproducibility
- Reviews can sting — **every** researcher gets rejections; top venues reject 75–85%
- Rejection ≠ bad work: revise using the reviews and resubmit elsewhere

8 RESPONDING TO REVIEWERS (REBUTTAL CRAFT)

- Golden rules:
- **Thank, never fight** — even unfair reviews get professional responses
- **Answer every point** — numbered, quoting the reviewer comment
- **Show the change:** "We added Section 4.3 (pp. 7–8) reporting the ablation requested"
- If you disagree: respectful evidence-based argument, not dismissal
- Reviewers are often **right about confusion** even when wrong about the fix — if they misunderstood, your writing caused it

9 CASE STUDY: ONE PAPER'S JOURNEY

- **Timeline of a real-style paper:**
- v1 → Conference A (CORE B): **reject** — "no comparison with transformer baselines"
- +6 weeks: added 2 baselines, ablation study
- v2 → Conference B: **accept with shepherding**; camera-ready after addressing meta-review
- +1 year: extended with new dataset (+40% material) → journal Q2: **major revision** → **accept**
- **Discussion:** what did the rejection contribute to the final quality? Map each reviewer demand to a paper improvement.

10 COPYRIGHT & OPEN ACCESS

- On acceptance you typically sign a **copyright transfer** (IEEE/ACM) or **license to publish**
- **Open access routes: Gold** (pay APC, free to readers) · **Green** (self-archive accepted manuscript in repository/arXiv per embargo rules) · Diamond (free both ways)
- Check **Sherpa Romeo** for a journal's self-archiving policy (https://openpolicyfinder.jisc.ac.uk/)
- You may normally post your **accepted version** on your homepage/institutional repository — read the agreement, don't guess

11 PEER-REVIEW WORKSHOP

- Today you become reviewers:
- Each group receives **another group's full draft** + a real review form: summary, strengths, weaknesses, detailed comments, score (1–5) on novelty/soundness/clarity/presentation
- (25 min) Review in pairs — be specific: point to sections, suggest fixes
- (10 min) Reviews returned; authors write a 3-point **mini-rebuttal plan**
- (5 min) Debrief: what did reading someone else's draft teach you about your own?
- **Due Week 12:** final 4–6 page IEEE-format mini-paper incorporating review feedback

12 WRAP-UP & NEXT WEEK

- **Key takeaways**
- Thesis = paper's deeper, pedagogical sibling — template compliance + chapter discipline + supervisor rhythm
- Dissemination is a pipeline: venue choice → peer review → rebuttal → camera-ready; rejection is feedback, not failure
- Check venues against predatory red flags **every time**
- **Next week (finale):** Presentation techniques + viva facing + **in-class mini-conference** — every group presents (10 min) + mock viva (5 min).